

Weak-star Delta-points Do Not Appear In Duals Of Subspaces Of Small Unconditional Basis Spaces

Automatic math-discovery smoke test

June 14, 2026

Source Problem

The source paper is Trond A. Abrahamsen, Ramon J. Aliaga, Vegard Lima, Andre Martiny, Yoel Perreau, Antonin Prochazka, and Triinu Veeorg, *Delta-points and their implications for the geometry of Banach spaces*, arXiv:2303.00511 [1]. The relevant question appears as Question 5.18 on page 23 of the arXiv PDF, after Corollary 5.17.

Also, note that if we view ℓ_1 as the dual of c , then ℓ_1 contains a weak* super Δ -point by Theorem 5.6. However, ℓ_1 as the dual of c_0 contains no weak* Δ -points by Corollary 5.15.

We have one final corollary of this subsection. Recall that $\mathfrak{D}$ -points can be lifted from subspaces.

Corollary 5.17. *If X is a subspace of a Banach space with a shrinking k -unconditional basis for $k < 2$, then X contains no $\mathfrak{D}$ -points.*

Let us note that there is no quotient version of Corollary 5.17 since every separable Banach space is a quotient of ℓ_1 . However, it is natural to ask the following.

Question 5.18. Can we conclude that X^* contains no weak* Δ -points if X is a subspace of a Banach space with a shrinking k -unconditional basis for $k < 2$?

According to Cowell and Kalton [CK10] a separable Banach space X has *property (au*)* if $\lim_{n \rightarrow \infty} \|x^* + x_n^*\| = \lim_{n \rightarrow \infty} \|x^* - x_n^*\|$ whenever $x^* \in X^*$, $(x_n^*)_{n \geq 1}$ is a weak*-null sequence and both limits exist. In particular, spaces with Kalton's property (M*) and spaces with a 1-unconditional finite dimensional decomposition satisfy this property.

It was shown in [CK10, Theorem 4.2] that a separable Banach space X has property (au*) if and only if for every $\varepsilon > 0$ we have that X is $(1 + \varepsilon)$ -isomorphic to a subspace of a Banach space Y with a shrinking 1-unconditional basis. This is in turn equivalent to the fact that for every $\varepsilon > 0$ we have that X is isometric to a subspace of a Banach space Y with a shrinking $(1 + \varepsilon)$ -unconditional basis. Hence by Corollary 5.17 separable

In text form, the question is:

Can we conclude that X^* contains no weak* Δ -points if X is a subspace of a Banach space with a shrinking k -unconditional basis for $k < 2$?

Result Type

This packet gives a **candidate full affirmative answer** to Question 5.18. It proves the statement as asked, with the weak-star topology on X^* coming from the predual X .

Definitions And Source Inputs

For a Banach space E , write B_E for its closed unit ball. A point $x^* \in S_{E^*}$ is a weak-star Δ -point if every weak-star slice of B_{E^*} containing x^* has points arbitrarily close to distance 2 from x^* .

It is a weak-star super Δ -point if the same condition holds for every relatively weak-star open subset of B_{E^*} containing x^* .

We use the following theorem from the source paper [1, Theorem 5.6]: if E^* contains a weak-star Δ -point, then E^* contains a weak-star super Δ -point.

We will also use the standard fact that if $T : E \rightarrow E$ is compact, then $T^* : (B_{E^*}, w^*) \rightarrow (E^*, \|\cdot\|)$ is continuous. This is the compact-adjoint continuity used in the proof of [1, Theorem 5.12].

Proof Intuition

The source paper already proves a no-weak-star- Δ theorem when the space itself has a shrinking k -unconditional basis. The only obstacle in Question 5.18 is that a subspace $X \subset Y$ need not inherit the finite-rank coordinate projections of Y . Passing to the dual removes this obstacle: X^* is the quotient of Y^* by restriction. A weak-star convergent net in the quotient unit ball can be lifted to the ambient dual unit ball, and weak-star compactness gives a convergent subnet. The ambient shrinking unconditional basis supplies a uniform distance bound strictly below 2 on that subnet. Since quotient distance is no larger than ambient distance, the same obstruction holds in X^* .

A Quotient Stability Lemma

Lemma 1. *Let E be a Banach space. Suppose there is a family $(T_\lambda)_{\lambda \in \Lambda}$ of compact operators on E such that*

$$r := \sup_{\lambda \in \Lambda} \|I_E - T_\lambda\| < 2 \quad \text{and} \quad T_\lambda^* x^* \rightarrow x^* \quad \text{in norm for every } x^* \in E^*.$$

Then there is a constant $c < 2$ such that every weak-star convergent net $(x_\alpha^) \subset B_{E^*}$, say $x_\alpha^* \rightarrow x^*$, has the following cofinal closeness property: for every index α_0 , there is $\alpha \succeq \alpha_0$ with*

$$\|x_\alpha^* - x^*\| \leq c.$$

Proof. Choose $\varepsilon > 0$ so that $c := r + 2\varepsilon < 2$. Fix a weak-star convergent net $(x_\alpha^*) \subset B_{E^*}$, with limit x^* , and fix an index α_0 . Pick λ such that

$$\|x^* - T_\lambda^* x^*\| \leq \varepsilon.$$

Since T_λ is compact, $T_\lambda^* x_\alpha^* \rightarrow T_\lambda^* x^*$ in norm. Hence there is an index β such that $\|T_\lambda^* (x_\alpha^* - x^*)\| \leq \varepsilon$ whenever $\alpha \succeq \beta$. Choose $\alpha \succeq \alpha_0, \beta$. Then

$$\begin{aligned} \|x_\alpha^* - x^*\| &\leq \|x^* - T_\lambda^* x^*\| + \|T_\lambda^* (x_\alpha^* - x^*)\| + \|x_\alpha^* - T_\lambda^* x_\alpha^*\| \\ &\leq \varepsilon + \varepsilon + \|I_E - T_\lambda\| \leq r + 2\varepsilon = c < 2. \end{aligned}$$

□

Lemma 2. *Let X be a closed subspace of a Banach space Y . Assume that Y satisfies the conclusion of Lemma 1; that is, there is $c < 2$ such that every weak-star convergent net in B_{Y^*} has a cofinal point within distance c of its limit. Then X^* has no weak-star super Δ -points.*

Proof. Let $Q : Y^* \rightarrow X^*$ be the restriction map. It is weak-star to weak-star continuous and, by Hahn–Banach, $Q(B_{Y^*}) = B_{X^*}$.

First we prove the same cofinal closeness property in B_{X^*} . Let $(u_\alpha^*) \subset B_{X^*}$ be a weak-star convergent net with limit u^* , and fix α_0 . Restrict to the tail $\{\alpha : \alpha \succeq \alpha_0\}$. For each α in this tail, choose $v_\alpha^* \in B_{Y^*}$ with $Qv_\alpha^* = u_\alpha^*$. By weak-star compactness of B_{Y^*} , the net (v_α^*) has a

weak-star convergent subnet (v_β^*) , with limit $v^* \in B_{Y^*}$. Continuity of Q gives $Qv^* = u^*$. By the assumed ambient property, there is a subnet index β such that

$$\|v_\beta^* - v^*\| \leq c.$$

Therefore

$$\|u_\beta^* - u^*\| = \|Q(v_\beta^* - v^*)\| \leq \|v_\beta^* - v^*\| \leq c.$$

This proves the cofinal closeness property in B_{X^*} .

Now suppose, for contradiction, that $u^* \in S_{X^*}$ is a weak-star super Δ -point. Direct the set of pairs (U, n) , where U is a relatively weak-star open neighborhood of u^* in B_{X^*} and $n \in \mathbb{N}$, by reverse inclusion in U and increasing n . Since u^* is weak-star super Δ , for every (U, n) we may choose $u_{U,n}^* \in U$ with

$$\|u_{U,n}^* - u^*\| > 2 - \frac{1}{n}.$$

This net converges weak-star to u^* , and eventually its distance from u^* is larger than c , because $c < 2$. This contradicts the cofinal closeness property just proved. Thus X^* has no weak-star super Δ -points. $\square$

Main Result

Theorem 1. *Let X be a closed subspace of a Banach space Y . If Y has a shrinking k -unconditional basis for some $k < 2$, then X^* , equipped with the weak-star topology induced by X , contains no weak-star Δ -points.*

Proof. Let (e_n) be the shrinking k -unconditional basis of Y , and let $P_n : Y \rightarrow Y$ be the n -th partial sum projection. The operators P_n are finite-rank, hence compact. Since the basis is shrinking, $P_n^* y^* \rightarrow y^*$ in norm for every $y^* \in Y^*$. Since the basis is k -unconditional, the tail projections satisfy

$$\|I_Y - P_n\| \leq k \quad \text{for every } n.$$

Thus Lemma 1 applies to $E = Y$, with $r \leq k < 2$. Lemma 2 then shows that X^* contains no weak-star super Δ -points.

Finally, by [1, Theorem 5.6], the existence of a weak-star Δ -point in X^* would imply the existence of a weak-star super Δ -point in X^* . This is impossible by the previous paragraph. Therefore X^* contains no weak-star Δ -points. $\square$

Corollary 1. *Question 5.18 of Abrahamsen–Aliaga–Lima–Martiny–Perreau–Prochazka–Veeorg has an affirmative answer.*

Verification Notes

The main point to verify is Lemma 2. The lift (v_α^*) is taken only on an arbitrary tail of the original net in B_{X^*} , so the weak-star compact subnet still corresponds to indices cofinal beyond the prescribed starting point. Since the restriction map $Q : Y^* \rightarrow X^*$ is weak-star continuous, the lifted subnet limit restricts to the original quotient limit. The quotient norm inequality $\|Qz^*\| \leq \|z^*\|$ then transfers the strict < 2 distance bound.

No numerical computation is used. The proof is independent of separability except for the standard countable-basis setting in the source question.

Novelty And Literature Check

Before promotion, the run indexes

`registry_index.tsv`, `solutions/index.tsv`, `attempts/index.tsv`, `proof_gaps/index.tsv`

were searched for arXiv:2303.00511, the title, and core phrases including Δ -points, Daugavet points, weak-star Δ -points, subspaces, and shrinking k -unconditional bases. No duplicate packet, prior attempt, or proof-gap entry was found.

A bounded web search on June 14, 2026 for exact and close variants of Question 5.18, including phrases such as “weak* Delta-points”, “subspace”, “shrinking k -unconditional basis”, and “2303.00511”, found the source paper but no separate later paper explicitly resolving this question. A local corpus search across the parsed arXiv sources and run target-pool indexes likewise found the source question and background mentions of the source paper, but no explicit later answer to Question 5.18.

Human Review Recommendation

Send for human review as a candidate full solution. The recommended review focus is the subnet/quotient step in Lemma 2; once that lemma is accepted, the specialization to shrinking k -unconditional bases is exactly the compact-approximation mechanism used in the source paper.

References

- [1] T. A. Abrahamsen, R. J. Aliaga, V. Lima, A. Martiny, Y. Perreau, A. Prochazka, and T. Veeorg, *Delta-points and their implications for the geometry of Banach spaces*, arXiv:2303.00511, 2023.
- [2] A. Bowers and N. J. Kalton, *An introductory course in functional analysis*, Universitext, Springer, 2014.